The Hydrogen Plant Problem

Residual CUSUM: adjust for operating conditions, then investigate.

Historical account, illustrative data

Syd Stewart’s account describes two hydrogen units in the 1970s. One used 17% more naphtha per tonne of hydrogen. Catalyst replacement brought its consumption back into line with the other unit. This supported a catalyst explanation, but did not establish the exact history of the damage.

Every chart and CSV in the accompanying article is generated data, not historical plant readings. The simulations demonstrate a method; they do not reconstruct or validate the events.

The useful measurement

Outlet methane also varies with load. Compare the actual reading with a validated healthy-operation model at comparable operating conditions. The residual is actual minus expected methane. Check instruments, feedstock, temperature, pressure and model fit before interpreting departures.

Two uses of CUSUM

A prospective monitoring CUSUM needs a fixed healthy reference, decision limits and an agreed response. Retrospective Bootstrap CUSUM charts centred on the whole-series mean fit possible stage boundaries. Their boundary dates do not establish when an online alarm would have sounded.

A whole-series-centred CUSUM returns approximately to zero by construction. That is not evidence of recovery. Fitted steps may approximate a gradual trend. Confidence settings are not probabilities of damage, online false-alarm rates or measures of severity.

Connect signals to action

Check the measurement and operating conditions, review steam and power events, and obtain engineering evidence. Related signals need not be independent. Validate false-alarm and detection performance locally, and avoid treating the illustrative formula or settings as plant operating rules.